PATRIZIA

INVESTMENT STRATEGY & RESEARCH

# Location analysis and pricing of amenities

## Portfolio construction in the era of data intelligence

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DATA INTELLIGENCE INSIGHT

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# 1 Executive Summary

**Urban development in European cities follows an increase in the supply of amenities.** The growth in amenities is lower for cities that have already gone through substantial densification over the last years.

**Modern location analysis means combining the knowledge of local experts and opinions based on artificial intelligence.** The PATRIZIA Amenities Magnet algorithm evaluates the attractiveness of a location based on the supply of amenities relative to the city. This is a benefit when understanding the drivers of a location, and benchmarking locations within the city.

**With cities evolving continuously over time, the PATRIZIA Amenities Magnet Dynamic measures the change of amenities across years.** It points out regions experiencing an upswing, and those that have become less attractive in terms of the amenities relative to the city.

**The well-being of an inhabitant is proportional to the amenities accessible to them within a 15-minute bicycle ride.** This is what is meant by the concept of "chrono-urbanism." The PATRIZIA Amenities Magnet 15 Minutes gives a disaggregated scoring to which degree a location will supply tenants with the seven basic urban needs: commuting, living, caring, working, educating, supplying and enjoying.

**The monetary value tenants place on their proximity to amenities varies, depending on the type of amenity.**

- For example, average rents in Manchester decrease by about -2% if the nearest tram stop is 200 m away, and drop to -4% to -6% at a distance of 1.5 km.
- The greater the distance to the nearest bar, the higher the rent discount. This value holds until 700 m when the effect flattens out.

**Between 2019 and 2021, rents in Manchester grew on average by ca. 3 to 4% p.a., with higher growth rates in the outer circles than in the city centre.** The highest increase in rents of almost 4% CAGR was observed for properties closer than 100 m to a bar. Regarding the distance to the nearest tram stop and ATM, the potential for rental growth increases with every additional metre away from these amenities.

**The investment strategies that PATRIZIA is pursuing in the era of data intelligence are based on machine learning techniques combined with explainable artificial intelligence methods.** These can identify value drivers and clusters of assets to consider in customised investment strategies.

# 2 Introduction

The supply of amenities is undoubtedly related to the value of real estate

Amenities reflect our well-being. This increasingly recognised fact is observable across urban areas worldwide. Amenities might be generally defined as "location-specific goods," but they involve much more than that. They can appear, for example, as restaurants across cities or as an ensemble of specific amenities inside a neighbourhood, such as schools, supermarkets and bus stations. Taken together, they are a crucial ingredient when determining whether a location is attractive or not. The supply of amenities is undoubtedly related to the value of real estate: tenants might be willing to pay more for assets close to value-enhancing amenities, saving them transport costs. The benchmarking of locations via amenities, and the quantification of their impact on rents, are the central research questions of this PATRIZIA Insight on Location analysis and pricing of amenities - portfolio construction in the era of data intelligence.

Before we examine the relationship between amenities and the value of real estate, several aspects should be explored.

- We need to identify the supply and contribution of amenities to the utility function of tenants. In other words, what type of amenities affect the well-being of tenants positively and which negatively?
- Rather than counting the number of amenities, an algorithmic benchmarking approach is necessary to measure the attractiveness of locations within cities using the urban amenities' footprint.
- An intelligent model will calculate the extent to which rents react positively or negatively to the availability of, and changes to, amenities in a residential market.

We believe that the understanding of locations and identification of investment opportunities are nowadays not only based on our local experience and footprint. We believe that technology, intelligent algorithms and machine learning methods can help us confirm our knowledge and, most importantly, advise us about unknown patterns and market forces that are difficult to obtain. The way PATRIZIA incorporates machine learning algorithms in the portfolio construction is described in this report.

# 3 Urban development through the goggles of digital information

> In the era of digitalisation, the amount of information collected via OpenStreetMap or Google Maps has drastically changed how we see and interpret urban areas

We can't count all the amenities in a city. This argument might be one of the main reasons why the analysis of amenities is not standardised and seen as difficult to carry out. However, in the era of digitalisation, the amount of information collected via OpenStreetMap or Google Maps has drastically changed how we see and interpret urban areas. Nowadays, we have access not only to the geographic location of a group of amenities but to the details inside each point of interest, such as commentaries, pictures and videos. While this large amount of information might be helpful for understanding areas, using all the details can lead to information overload and biased signals. For real estate investments, not all points of interest are relevant when the aim is to identify spatial patterns that lead to additional knowledge.

When merging the socioeconomic database of Oxford Economics with data from OpenStreetMap and Google Maps for 31 European cities in 2017 and 2021, we can see the relationship between the supply of amenities and urban development.[1,2,3] First, the density of households is highly correlated (>88%) with the density of amenities per km$^2$. As of 2021, there were 112 amenities available for 1,111 households per km$^2$ on average. Low-densified cities, such as Helsinki , Lisbon, Lyon and Stockholm, supply their fewer than 500 households per km$^2$ with fewer than 50 amenities per km$^2$, while high-densified cities, such as Berlin, Brussels, Copenhagen or Munich, provide on average more than 270 amenities to more than 2,000 households per km$^2$. The cities of Warsaw and Dublin are worth mentioning as they seem to be under- and over supplied with amenities relative to all observed cities, respectively. Overall, the supply of amenities per km$^2$ increases by one unit for every additional ten households, ceteris paribus.

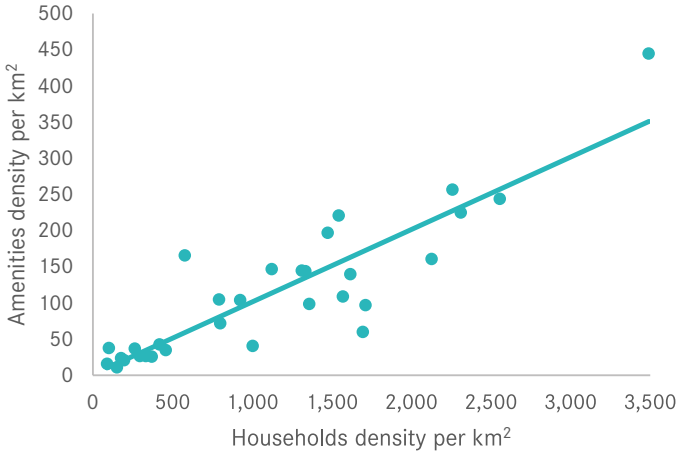
$$\frac{\widehat{\#Amenities}}{km^2} = 1.351 + 0.100 * \frac{\widehat{\#Households}}{km^2}$$

[1] We exclude London and Paris because they are outliers in terms of size and proportion of supply amenities.

[2] The area of the cities is defined by the NUTS3 polygon from Eurostat.

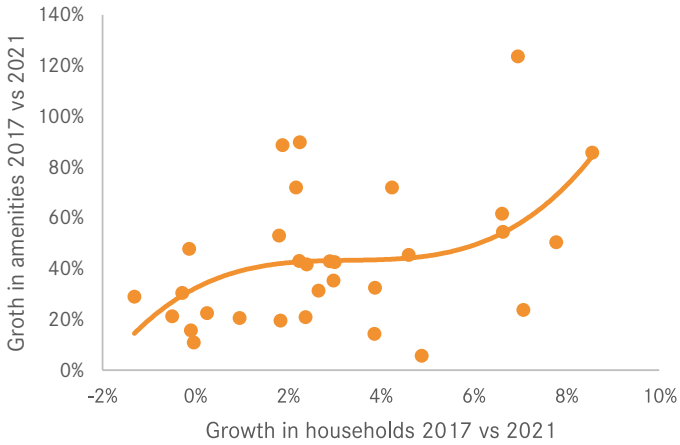
[3] OSM data collected as of February excluding ways.

Figure 1: Households and amenities density 2021 ($R^2$=78%)




Secondly, when measuring the change in the supply of amenities as a function of the change in households and household density as of 2021, in none of the observed cities did the supply of amenities decrease. In fact, the number of amenities in all cities grew from 2017 to 2021 by 43% on average. This is positively associated with the growth in the number of households. The lowest increase in both number of households (0.1%) and amenities (22%) was observed in the top seven German residential markets. Barcelona, Helsinki, Madrid, Luxembourg and Stockholm showed a substantial increase in amenities by ca. 90%, together with an increase in the number of households by 5.3% on average. Overall, a one percentage point growth in the number of households during these four years led to an expansion in the supply of amenities of 2.9 percentage points, ceteris paribus. In other words, demographic growth is accompanied by an increase in the supply of amenities.

Figure 2: Change in amenities and number of households ($R^2$=23%)



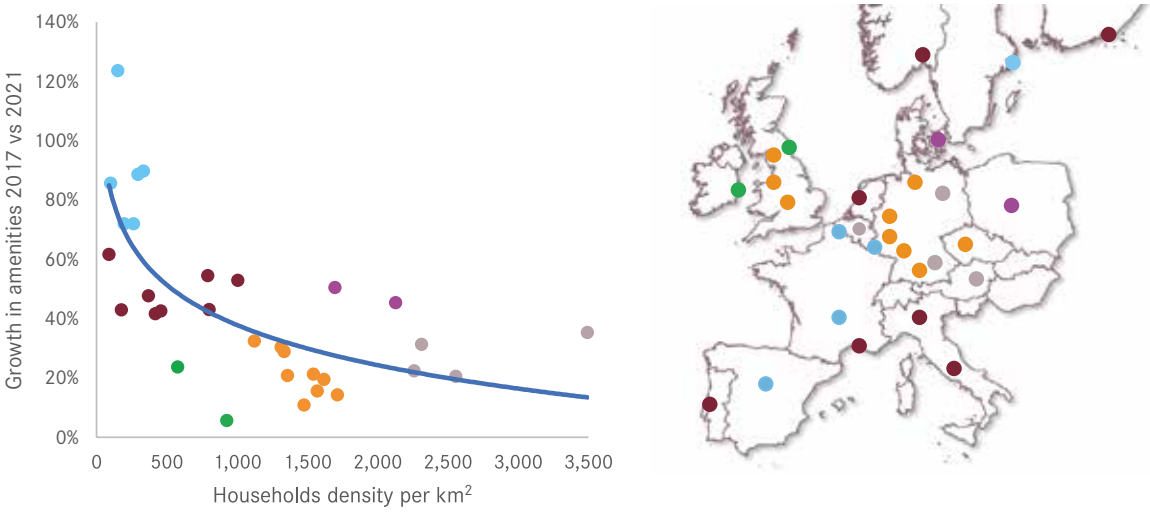

Source: PATRIZIA SE, Oxford Economics, OSM, Google

Growth in amenities is precisely limited for cities that have already seen substantial densification over the last years

Finally, the increase in amenities was more pronounced in cities with low household density per km² compared to high-densified cities. This means that the opportunity to change the amenities' urban fabric is directly restricted by household density. For example, Brussels, one of the most densely populated cities in Europe with almost 3,500 households per km², showed an expansion in the supply of amenities of roughly 35%. The opposite development was observed in Madrid, where amenities grew by almost 90% in an area with a density of at least 300 households per km². Overall, every additional 100 households per km² decreased the growth of amenities by 1.6 percentage points over the observed period, ceteris paribus.

$$\Delta \widehat{Amenities} = 0.523 + 2.865 * \Delta \widehat{households} - 0.160 * \frac{\widehat{Households\ tsd.}}{km^2}$$

Figure 3: Change in amenities and household density ($R^2$=53%)




Source: PATRIZIA SE, Oxford Economics, OSM, Google, Eurostat.

So far, our analysis reveals that the demographic expansion of European cities follows an increase in the supply of amenities that affect the well-being of households overall. Obviously, the larger the city in terms of households, the higher the supply of amenities per km². However, results show that the growth in amenities is precisely limited for cities that have already seen substantial densification over the last years. This observation allows us to conclude that the data captured through digital information systems such as OpenStreetMap and Google Maps correlates with and is explainable through official socioeconomic data, such as data from Oxford Economics. The opportunity to shape a specific city area in terms of (new) urban amenities is higher if a city is less densified.

# 4 Transforming noise to signals

The PATRIZIA Amenities Magnet evaluates the attractiveness of locations by measuring the quality-adjusted and location-adjusted supply of amenities

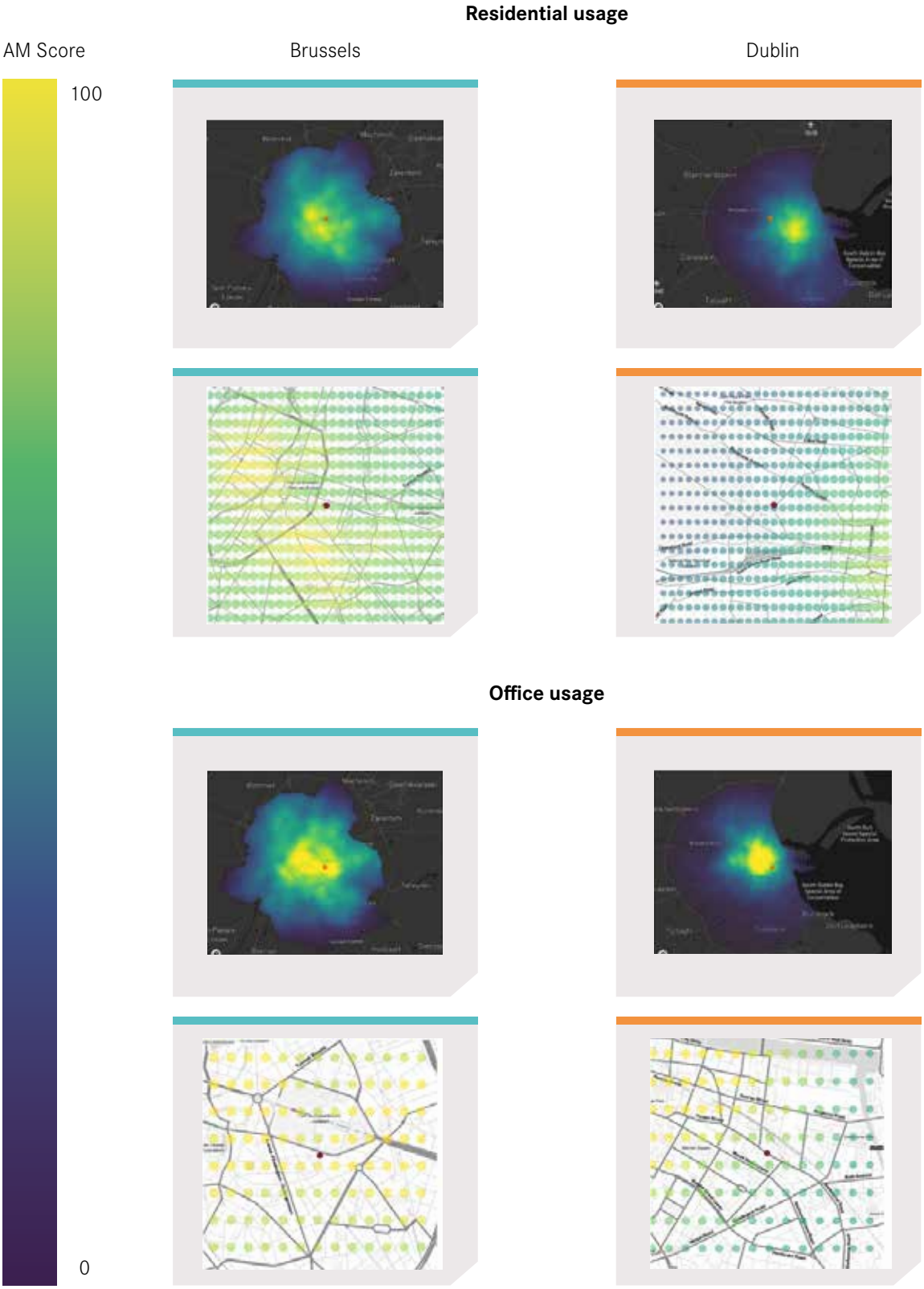
Unlike the market for other goods like labour or steel, there is no market for amenities: they are said to be location-specific non-excludable goods, that is, they are consumed once they are accessible. They are not fungible, so their price is not defined by supply or demand. Another critical aspect of amenities is that they affect households' satisfaction and firms' profits independently of their location choice. For example, a kindergarten or bus stop in a specific neighbourhood can positively contribute to the well-being of households, firms and the community itself, whether or not they are being consumed. That said, the marginal benefit of these amenities decreases if additional kindergartens and bus stops suddenly appear. Although this sudden increase in supply leads to an appreciation of the neighbourhood overall, the value of each single amenity decreases because households can access more kindergartens or bus stops. Therefore, the higher the substitution rate between amenities within a certain area, the lower the marginal value of amenities to households or firms will be.

The latter theoretical description of the specific definition of amenities is necessary to create a valid benchmarking tool that aims at precisely measuring the quality-adjusted and location-adjusted supply of amenities. This tool is called the PATRIZIA amenities magnet algorithm, and it makes use of unsupervised machine learning (ML) methods with two fundamental assumptions. Firstly, since amenities are defined as location-specific goods, either public or private, an approach that evaluates the attractiveness of locations by simply counting the number of amenities will always be biased. This seems evident for cases in which households have access, for example, to several swimming pools within a radius of 500 m, so that the marginal benefit of households for any additional swimming pool decreases. Secondly, not all amenities influence the well-being of households and the profits of firms homogenously. While this is a solid assumption, it states that the location choice of tenants includes and excludes certain amenities.

Based on the points of interest (POI) database of OpenStreetMap and Google Maps, the PATRIZIA amenities magnet score benchmarks a location within a city on a regular grid and provides score values normalised to the interval 0 to 100, with 100 indicating the best location with the highest supply of amenities. Quantile values are provided to account for the skewness of the distribution of the scores.

When benchmarking Brussels and Dublin for residential and office investment, the results of the PATRIZIA amenities magnet analysis are shown in a heatmap. A stronger yellow shows a higher score and a better supply of amenities; purple indicates low scores and low location quality due to the neighbourhood's lack of relevant amenities. In the Brussels and Dublin city centres, scores are higher than in outer region. Each target location is marked as a coloured orange dot.

## Figure 4: PATRIZIA Amenities Magnet for residential and office assets




Source: PATRIZIA SE

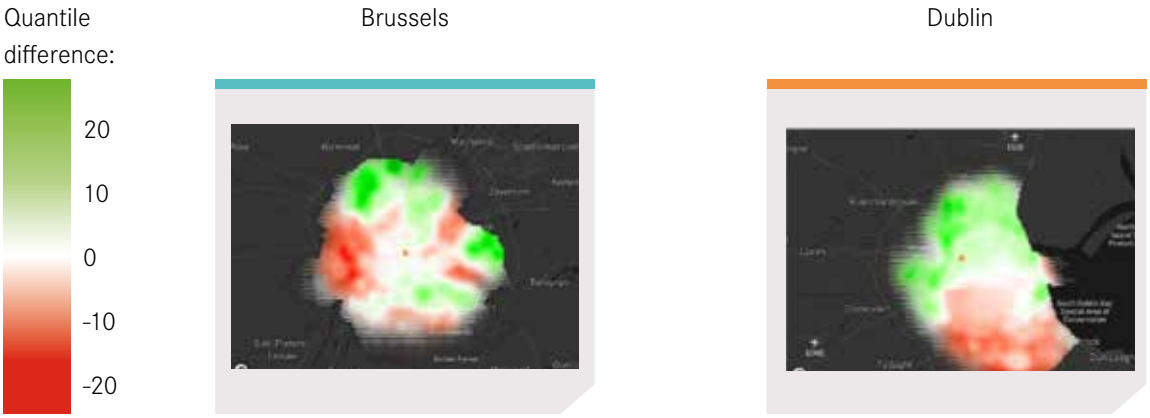
The residential (office) locations in Brussels reach a score of 83 (96) out of 100 points, which corresponds to the 97% quantile. Therefore, the locations of the assets belong to the top 3% of residential (office) locations within the city of Brussels. Both assets are located between two adjacent magnets – that, yellow regions – and are therefore in a neighbourhood that is well covered regarding amenities. The residential (office) location in Dublin reaches a score of 52 (77) out of 100 points, which corresponds to the 88% (95%) quantile. Thus, the location belongs to the top 12% (5%) of residential (office) locations within the city of Dublin. Although the office asset is located at the edge of the yellow magnet, it still belongs to the top quartile of locations within the benchmark area.

> The PATRIZIA Amenities Magnet Dynamic allows to compare the amenity-wise development of regions over time

As an additional algorithm, the PATRIZIA amenities magnet dynamic calculates both historical and current scores and quantile values based on the same algorithm and the POI databases of 2017 and 2021. It allows us to compare the development of amenities over the last four years and thus gives an impression of which parts of a city experienced an upswing and which have lost attractiveness regarding their amenities, relative to the rest. Results for assets in Brussels and Dublin are also shown as a heatmap. The colour code of the heatmap refers to the change in quantiles in 2021 compared to 2017. White areas represent solid locations which stayed the same and therefore developed at the average city level. Green areas experienced a faster positive development during the past four years and red ones the contrary.

The location in Brussels maintained a solid quantile position (2017: 95%; 2021: 97%), comparable to the entire centre of the city. In terms of amenities, Dublin developed exceptionally well in the northern part of the city. The residential location under consideration participated in this upswing and experienced an increase of 5 position points (2017: 83%; 2021: 88%).

Figure 5: PATRIZIA Amenities Magnet Dynamic for residential assets




Source: PATRIZIA SE

# 5 Adjusting signals into knowledge

> “Chrono-urbanism” differentiates between seven well-being factors that affect specific aspects of urban life: commuting, enjoying, supplying, educating, working, caring and living

The PATRIZIA amenities magnet score presented in the last section evaluated the attractiveness of a location within a city based on the supply of amenities. This score enables the benchmarking of a city in terms of strong and weak magnets across different asset classes. However, it is unlikely that an inhabitant will care about all the amenities in the city when deciding where to live. Instead, they might be interested in the supply and quality of the amenities available in their desired community. This concept is called the “15-minute readiness of cities,” and it postulates that societal well-being across the city is proportional to the supply of amenities and the walkability or cyclability in each living community. In other words, inhabitants need to be able to access all their basic living needs within less than 15 minutes on foot or by bicycle. The closer these amenities are to the tenant, the higher their urban welfare will be. We, therefore, adjust the PATRIZIA amenities magnet score to measure these selected welfare signals, and call the resulting analysis the PATRIZIA Amenities Magnet 15 Minutes.

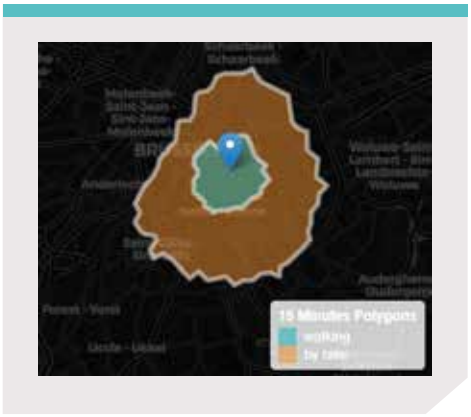
“Chrono-urbanism” differentiates between seven well-being factors that affect specific aspects of urban life: commuting, enjoying, supplying, educating, working, caring and living. While each of these factors includes specific types of amenities inside the specific catchment area, the PATRIZIA Amenities Magnet 15 Minutes enables immediate benchmarking between cities across Europe. In other words, it benchmarks the supply of amenities an inhabitant can access in a normal catchment area.

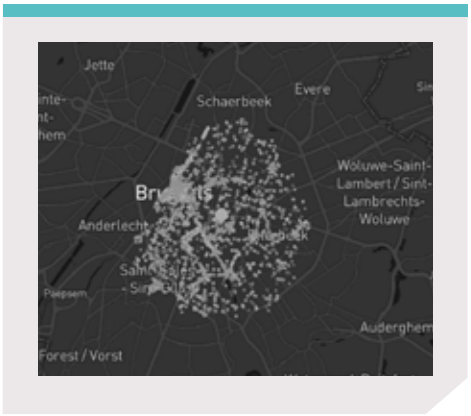
The dictionary of the PATRIZIA Amenities Magnet 15 Minutes assigns the relevant amenities of the POI database to the seven basic living needs categories. For the target locations in Brussels and Dublin, the polygons at the top of Figure 6 show the 15-minute catchment areas of the cities, on foot and by bicycle, respectively. The latter one is used as the benchmark area. The graphs at the bottom of Figure 6 visualise the scoring for all seven basic need categories as a spider diagram. They include the average scores, the 25% and the 75% quantiles, and the location scores.

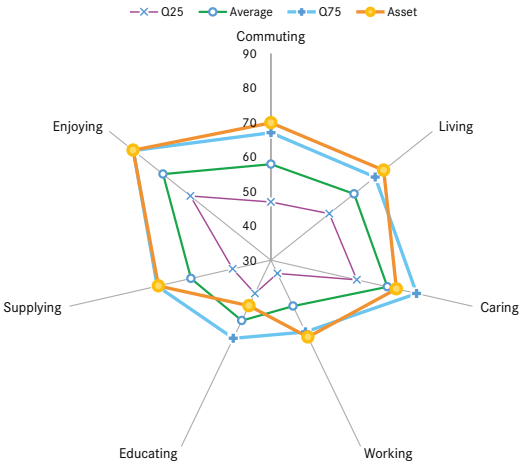
The residential location in Brussels belongs to the upper quarter in most basic living needs categories, except for educating and caring. This means that schools, universities and/or libraries, as well as care facilities such as pharmacies or doctors, apparently are supplied better in other parts of the benchmark area. In contrast, for Dublin the results show that for residential purposes the location doesn't belong to the upper quarter for any basic living need category except for supplying. Location scores are in between the average scores and the 75% quantiles. Even for supplying, the score is below the respective average score. This suggests that, according to the 15-minute readiness principle, there might be residential locations that are more suitable for residential purposes.

## Figure 6: PATRIZIA Amenities Magnet 15 Minutes for residential assets

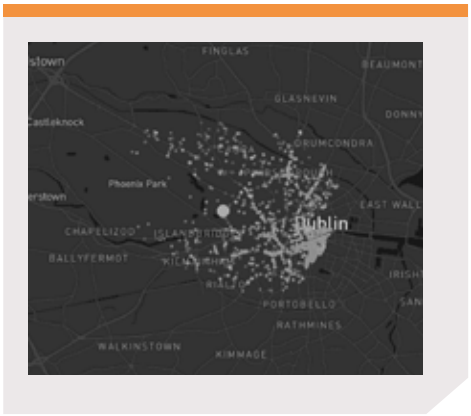
Brussels

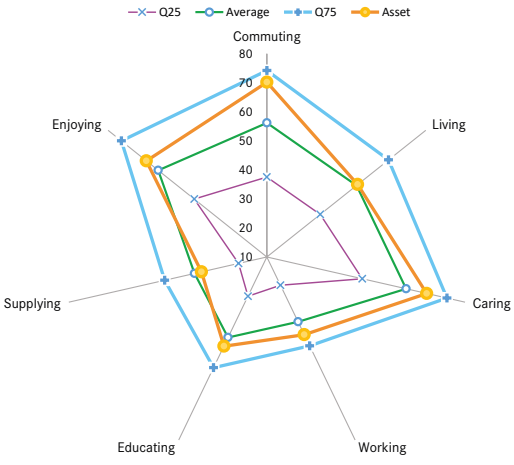



Dublin

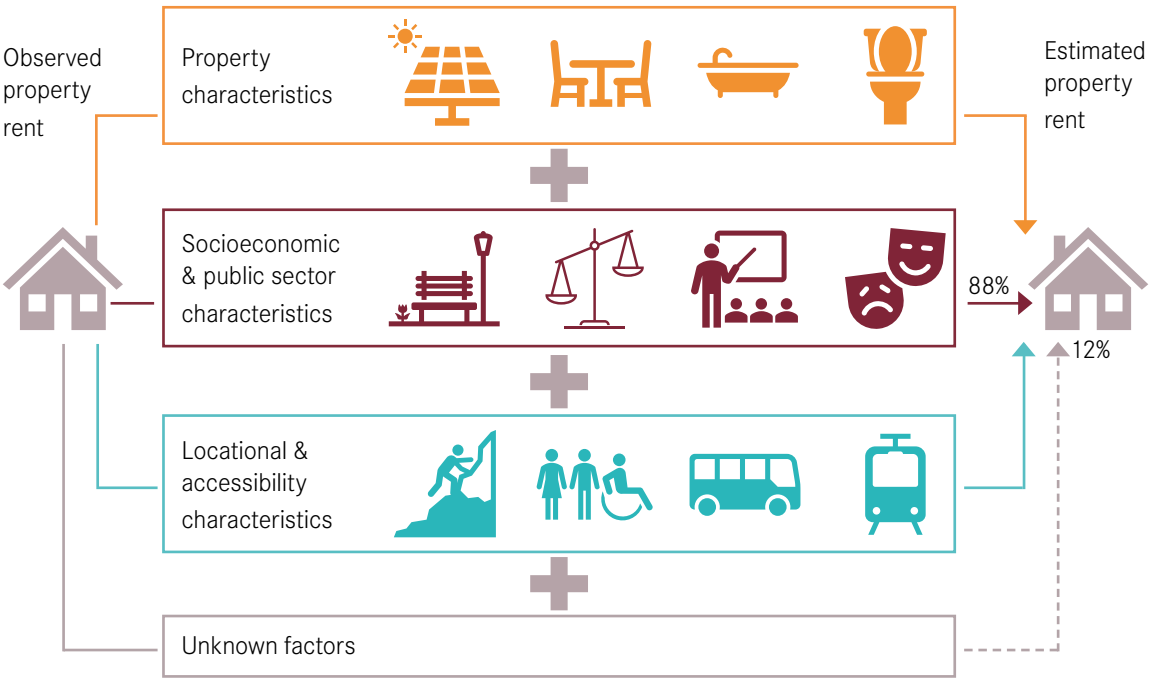


# 6 Connecting signals to prices

So far, amenities magnet algorithms have focused on measuring the attractiveness of locations based solely on the supply of amenities. However, to quantify whether the proximity to amenities affects a tenant's choice of location, we need to focus on rents. This means applying Rosen's (1974) hedonic framework that describes our willingness to pay for living. Basically, the hedonic approach equals the sum of the single contributions of characteristics that together explain a potential tenant's rent level. Conventionally, the factors affecting the level of rents include (i) property characteristics, (ii) local socioeconomic and public sector characteristics and (iii) locational and accessibility characteristics.

Figure 7: Factors affecting residential rents and model accuracy




Source: PATRIZIA SE.

Driven by the increase in computational power over the last three decades, machine learning methods have become part of everyday life

Instead of applying the traditional linear hedonic framework, we adapt a machine learning algorithm to adjust rents according to the distance to the closest amenities. Driven by the increase in computational power over the last three decades, machine learning methods have become part of everyday life, examples being personalised web ads or mail spam filters. However, these methods have been strongly criticised for being opaque – a black box – when it comes to understanding the steps that lead to an outcome. Consequently, over the last decades, computational statistics has focused on methods to improve the interpretability of such complex models. This field is called explainable artificial intelligence (often referred to as XAI).

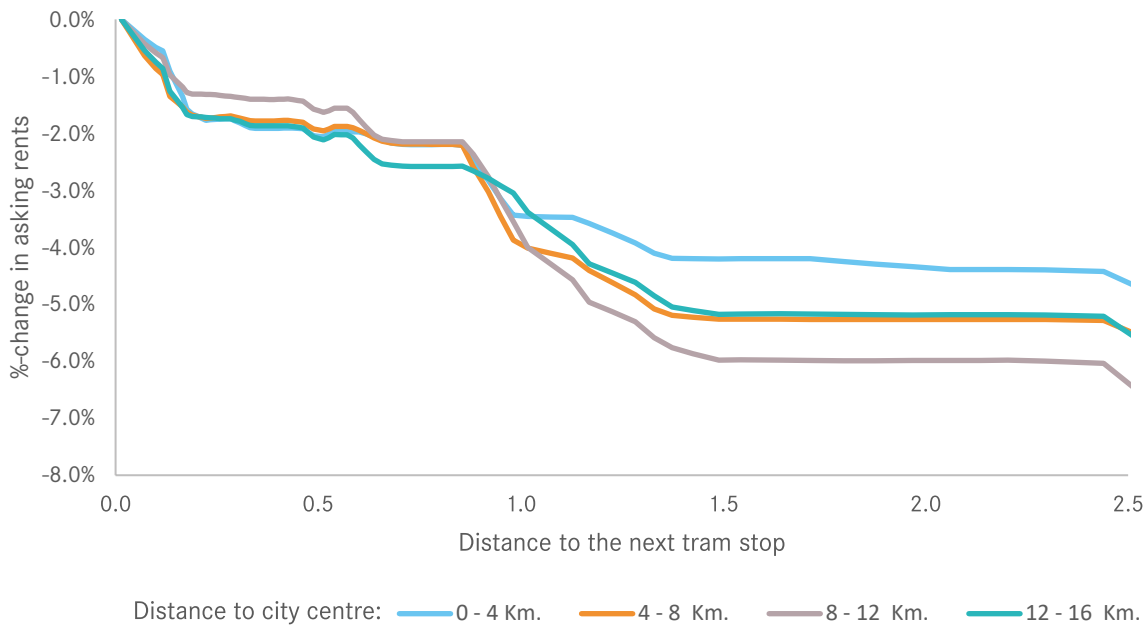
## Pricing of amenities across the city

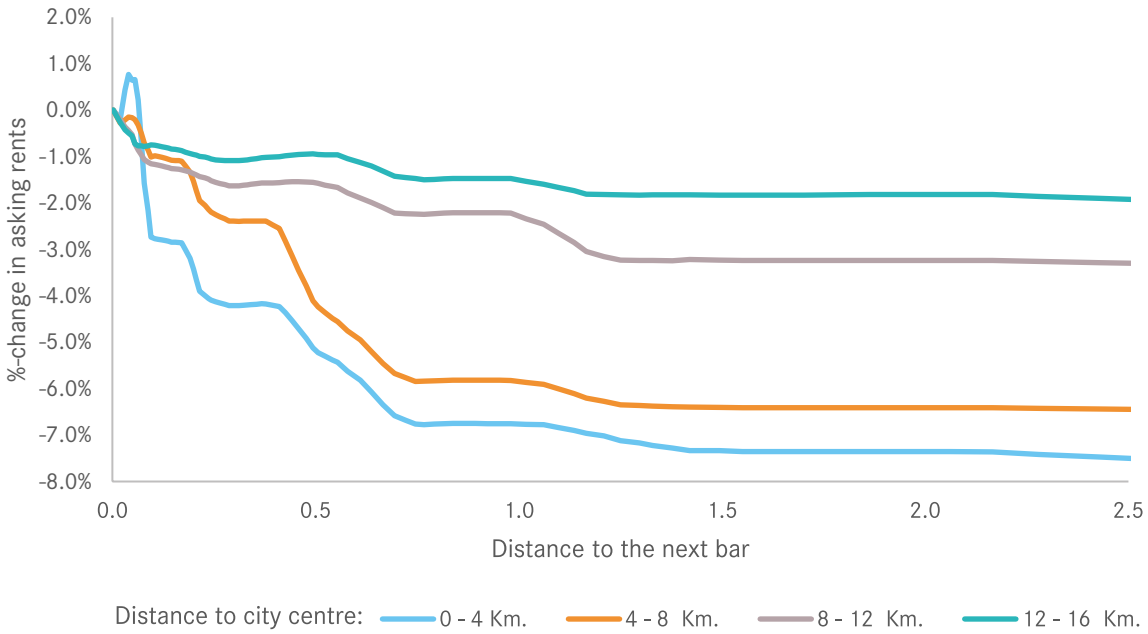
In a case study researching rents in Manchester, we apply the machine learning algorithm to a dataset containing 38k residential listings and a direct connection to OpenStreetMap and Google Maps. This means that the distances for each apartment to amenities have been calculated and are included as explanatory variables. Our final model has been optimised as an extreme gradient boosting (XGB) and reaches an explanatory power of 88%. This model allows us to understand the behaviour of rents in the context of the distance to selected amenities.[4]

We divided Manchester into four rings, each one four km to the city centre – that is, 0 to 4 km, 4 to 8 km, 8 to 12 km and 12 to 16 km. The explainable artificial intelligence results show that while tenants consider the proximity to amenities when they decide on a location, the monetary value they attribute to each amenity is anything but identical. The proximity to tram stops and bars has a significant influence on the level of rent. For example, the average asking rent of apartments decreases by ca. -2% if the nearest tram stop is 200 m away or more. While this effect applies to the entire market, regardless of the distance to the city centre, asking rents for apartments within 8 to 12 km to the city centre, with a tram stop further away than 1.1 km, decrease by -6%. In other words, every additional metre that takes the tenant further away from the tram stop reduces the potential income of the asset. This effect is strongest after 1 km, ceteris paribus. The proximity to a bar also pays off, especially for assets in the city centre: the further away the nearest bar is, the higher the rent discount, ceteris paribus. Interestingly, the willingness to pay for the proximity to a bar decreases the further away assets are from the city centre.

[4] To avoid a biased model that fails to predict unseen data (out-of-sample), the sample is divided into test and training set, with the model learning from the training set and being evaluated using the test set.

Figure 8: Change in rents according to the distance to the next tram stop and bar

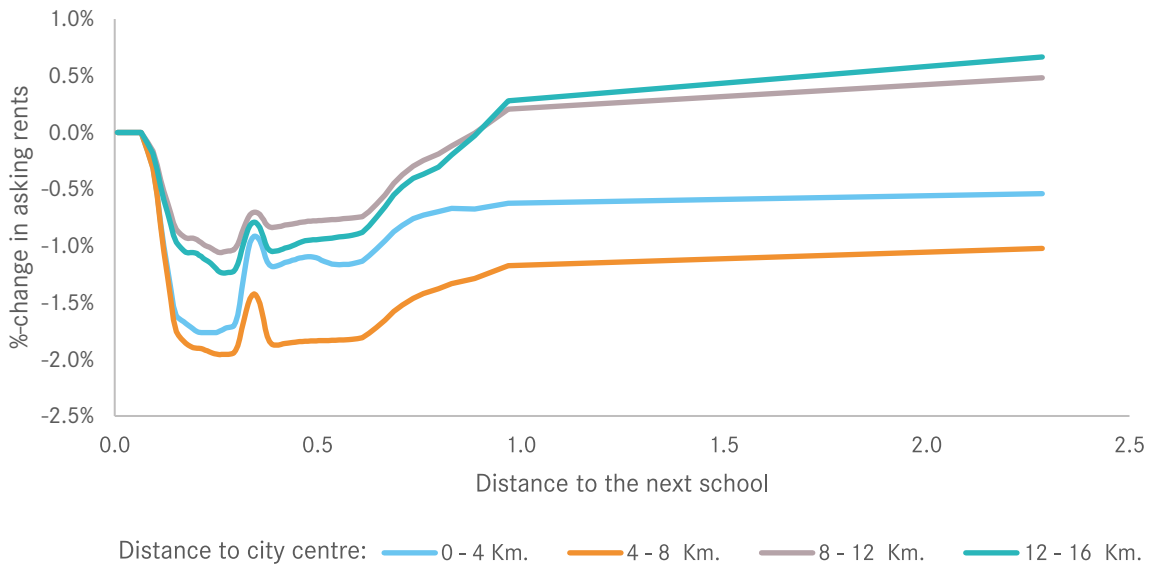


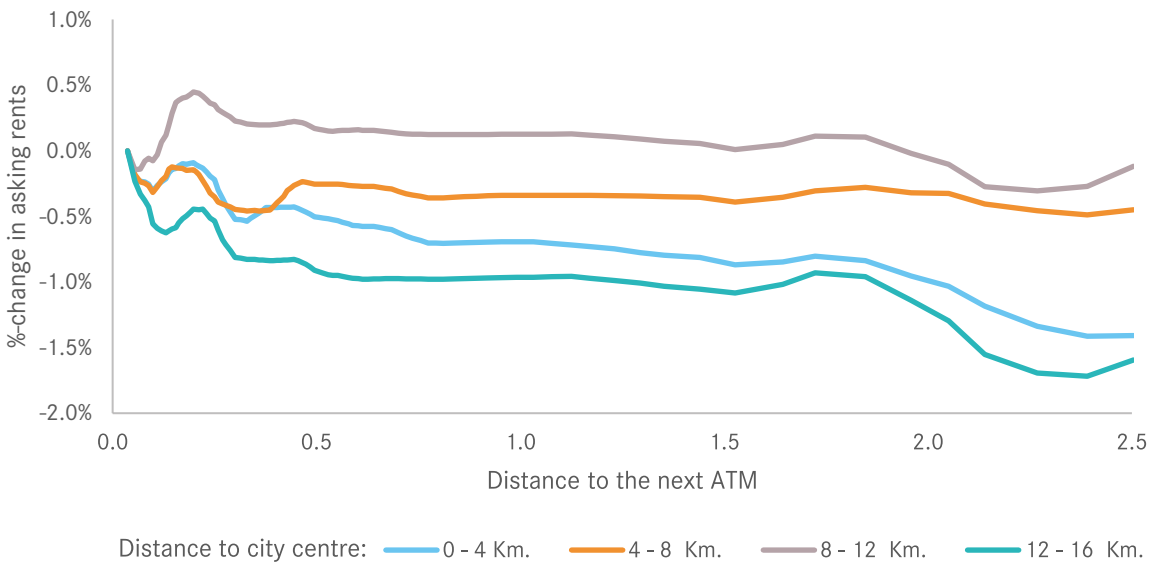




Source: PATRIZIA SE. The plots correspond to partial dependency plots.

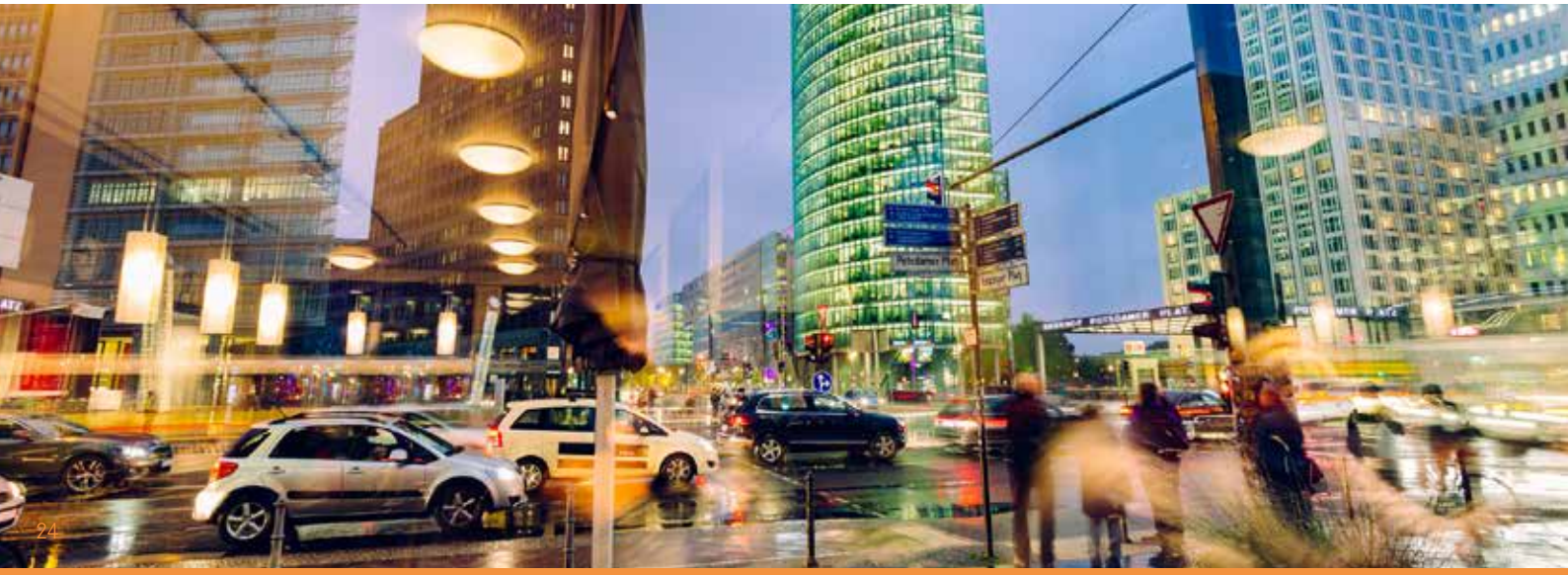
The willingness to pay for proximity to schools and ATMs is mixed. Independent of the distance to the city centre, a residential asset located between 200 and ca. 800 m away from a public school shows lower asking rents on average, ceteris paribus. The highest discount of -2% can be found for assets located 4 to 8 km away from the city centre. Interestingly, the discount reduces after 800 m, making a U-turn. The willingness to pay to live somewhere relative to the distance to an ATM is negative, meaning tenants are not willing to pay more for assets close to an ATM. However, the machine learning results have identified that the potential asking rent increases by almost 50 basis points if an ATM is close to assets within 8 to 12 km of the city centre.

Figure 9: Change in rents according to distance to nearest school and ATM



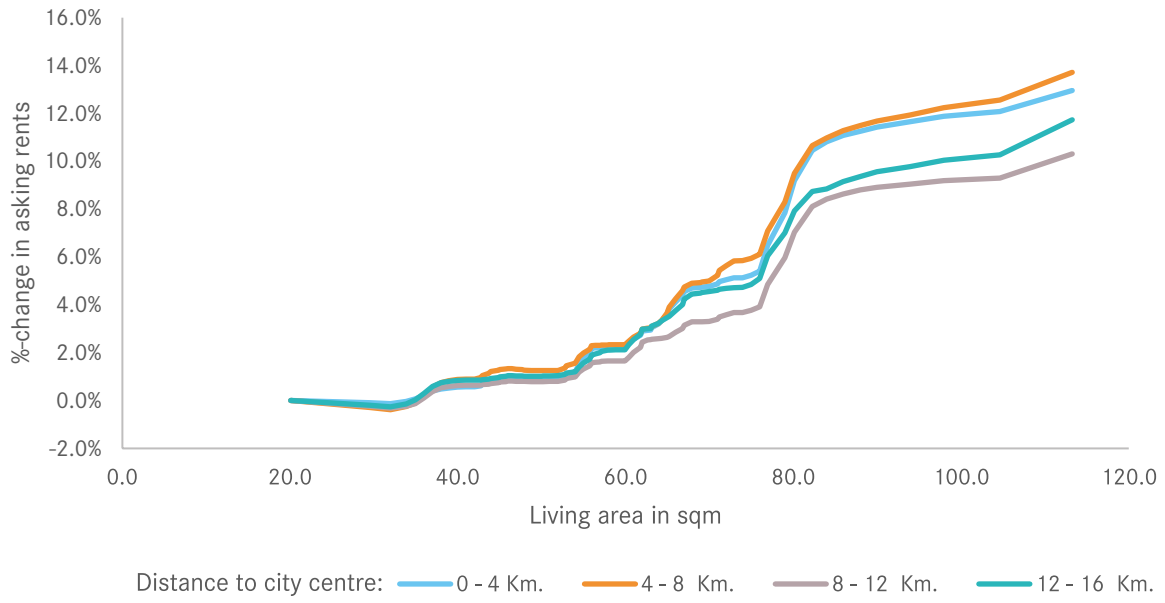



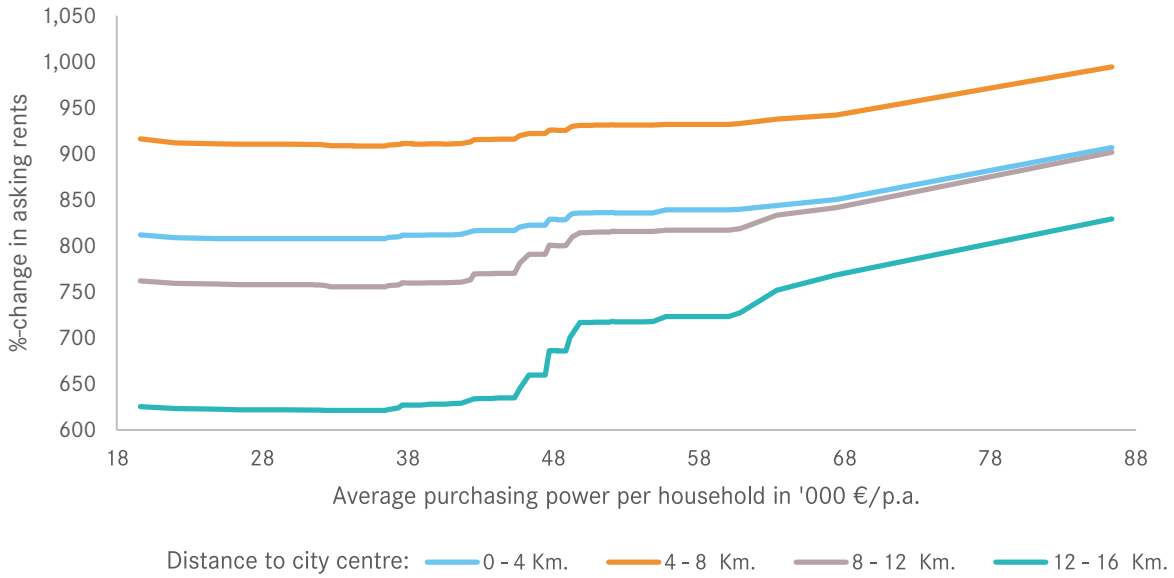

Source: PATRIZIA SE. The plots correspond to partial dependency plots.



Finally, the relationship between asking rents, household purchasing power and apartment size is examined. Regarding apartment size, the results show that the larger the apartment, the higher the monthly rent will be. Despite being obvious, the results show a significant increase in average rents for dwellings larger than 75 $m^2$ of almost four percentage points. The most substantial increase is 8 km away from the city centre. These results confirm that the potential for higher rents is more limited for assets larger than 90 $m^2$, and that the highest average rents are achieved within 4 to 8 km from the city centre. Regarding the purchasing power per household, average asking rents remain constant in areas where households have a purchasing power below ca. €45k p.a. There is a substantial rent premium for any increase in purchasing power, especially outside the city centre, that is, for properties 12 to 16 km away.

Figure 10: Change in rents according to living area and purchasing power



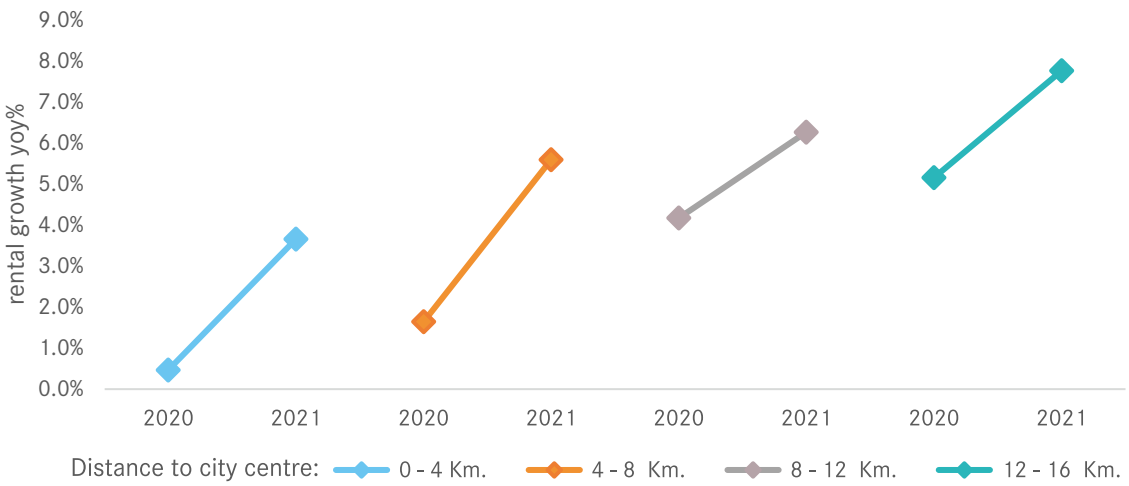




Source: PATRIZIA SE. The plots correspond to partial dependency plots.

## Pricing of amenities over time

Capturing the drivers of rental growth in a city can be challenging because they result from property characteristics, socioeconomic developments and public forces, or a combination of all these. One of the main advantages of machine learning algorithms is that they recognise the patterns and forces in data that affect rents. In Manchester, for example, the model identifies that between 2019 and 2021 rents grew on average by ca. 3 to 4% p.a. However, rental growth is proportionally related to the distance to the city centre. Between 2019 and 2020, asking rents grew on average by 0.5 to 1.5% for properties 4 to 8 km from the centre and by 4.0 to 5.0% if they were 12 to 16 km away. The differences were even more pronounced in 2021, when rents in the outer ring of Manchester grew almost twice as fast as rents in the city centre.

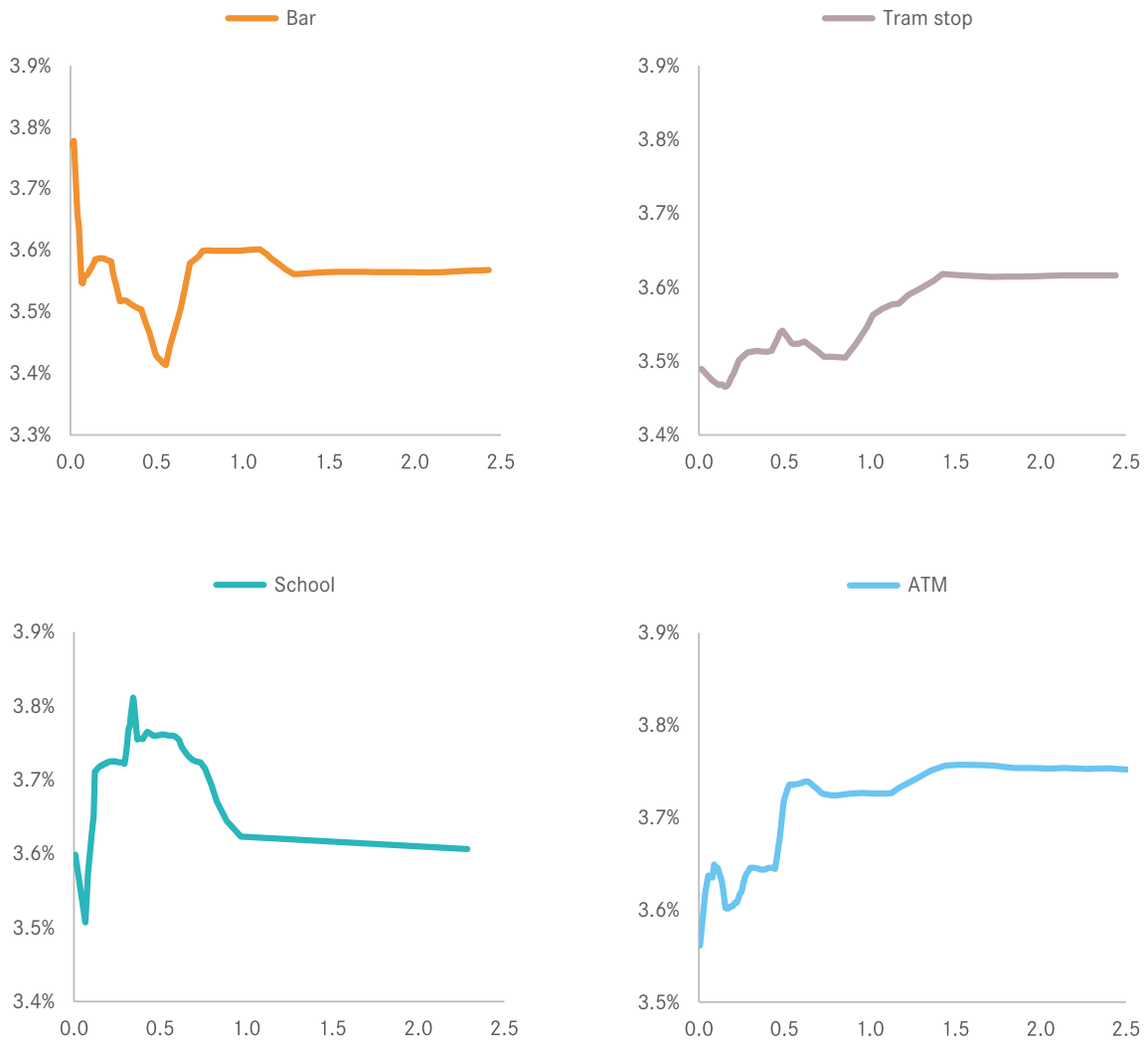
Figure 11: Growth of rents over time and according to distance to city centre




Source: PATRIZIA SE.

The distance to amenities affects the rental growth of residential assets. The highest increase in rents of 3.8% CAGR could be observed in assets closer than 100 m to the next bar, while rents of assets between 500 to 600 m away from the next bar grew 33 bps slower. Interestingly, the potential to increase rent rises with every extra metre of distance away from the next tram stop and ATM. The opposite is the case when looking at the rental CAGR of assets located 100 to 600 m away from the next public school.

Figure 12: CAGR of rents 2019–2021 according to distance to selected amenities




Source: PATRIZIA SE. The plots correspond to partial dependency plots.

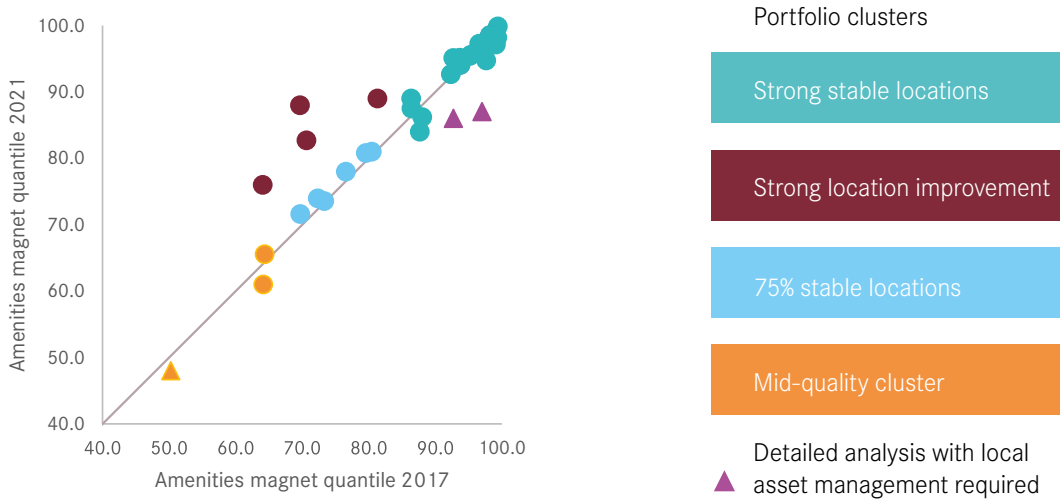
Machine learning algorithms recognise the patterns and forces in data that affect rents

# 7 Investment strategies in the era of data intelligence

Investment strategies in direct real estate nowadays involve a long-term-oriented view of the expected value drivers of rents and capital values. The knowledge acquired at PATRIZIA enables in-depth identification of signals to precisely understand current local drivers in terms of amenities and pricing. A value-boosting strategy occurs horizontally when the location is benchmarked relative to the city; it occurs vertically when the development of the location over time is identified. Moreover, the amenities magnet 15 quantifies whether the location fulfils its expected practice of supplying the tenant with those amenities they demand. Finally, the position of the asset relative to the existing amenities' urban fabric explains the level of and potential growth of rents. The machine learning approach identified, for example, that the distance to a public school leads to lower asking rents and confirmed that the potential for rental growth here is the strongest. All these considerations, methods and results build up the PATRIZIA Investment strategy.

Supervised and unsupervised machine learning techniques, together with explainable artificial intelligence, are not yet standard analytical instruments in the real estate industry. This is because their implementation and deployment require deep knowledge and experience, especially because the expected outcome is difficult to quantify without a solid analytical track record. However, once the analytical infrastructure has been built, an existing portfolio's analysis and risk assessment require only a few hours. This means that the identification of clusters and development of an action plan for an existing portfolio are now within our sights.

Figure 13: Investment strategies in the era of data intelligence




Source: PATRIZIA SE.

CASE STUDY

# Seville: assessing and benchmarking two locations

The Amenities Magnet Reports, unsupervised machine learning algorithms that benchmark locations based on the supply of amenities, help PATRIZIA understand where tenants can have access to which kind of  amenities. In this study, we look at 'El Porvenir' and 'Los Remedios, two distinctive areas of Seville.

Results of the PATRIZIA Amenities Magnet 15 Minutes indicate the mix of amenities that tenants can expect at a location. These can be used to derive a suitable tenant profile or an asset type recommendation.

**At El Porvenir, we observe:**

- The target location supplies tenants with amenities for education (Q99) and gives access to parks and playgrounds (Q83) (living/amenities for residential purposes)
- The target location lacks access to amenities for daily supplying (Q40) and caring (Q44).
- Likely tenant profile: families with (school-aged) children

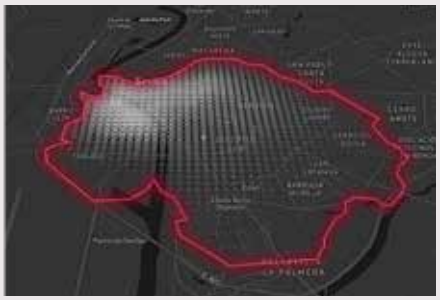
**While at Los Remedios we see:**

- The target location supplies tenants with amenities for enjoying (Q77), daily supplying (Q76), caring (Q77) and gives for example access to parks and playgrounds (Q89) (living/amenities for residential purposes).
- The target location lacks educational amenities in the nearer neighbourhood.
- Likely tenant profile: families with younger children and/or best-age-living

Unsupervised machine learning algorithms benchmark locations based on the supply of amenities

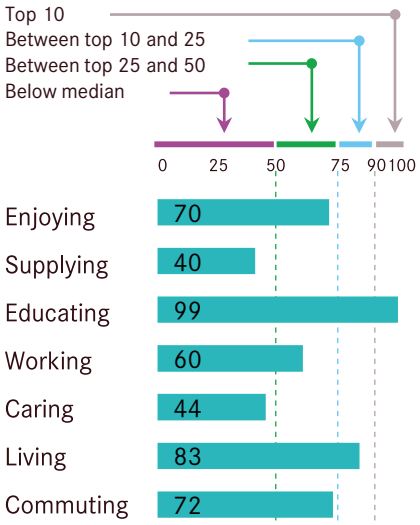
## What type of amenities mix can tenants expect?

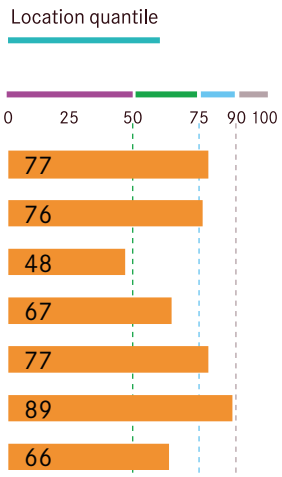
The following illustrations show how well the two locations in 'El Porvenir' and 'Los Remedios' are supplied with amenity groups relative to all locations that can be reached within 15 minutes by car.



**El Porvenir**
Seville - Spain

**Los Remedios**
Seville - Spain




| Families with (school-aged) children | Families with younger children and/or best-age-living |
|---|---|
| Location supplies tenants with education and access to amenities for residential purposes (living). Location lacks of access to daily supplying and caring. | Location supplies tenants with enjoying, daily supplying and caring and access to amenities for residential purposes (living). Location lacks of education. |

Source: PATRIZIA SE.

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